Stat 415 Lab 3 - Recommender Systems Pt. 2

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1 Introduction

In this project, we extend our exploration of recommender systems by focusing on collaborative filtering and predictive modeling techniques, with a particular emphasis on linear modeling. Our analysis is based on a rich dataset of restaurant reviews and demographic information collected in Evanston, Illinois. By leveraging user-level attributes and restaurant characteristics, we aim to uncover patterns in consumer preferences and develop personalized recommendation strategies.

This work has practical value for various audiences, including local residents, newcomers to the area, and Northwestern University students seeking dining options that align with their tastes and lifestyle. Through our modeling and analysis, we hope to enhance the restaurant discovery experience by providing informed, data-driven suggestions.

2 Exploratory Data Analysis

The raw data was collected from unspecified websites and contain 1500 reviews of real restaurants in Evanston, IL that people have written in the past. We used the variables:

- Reviewer Name — Name of the person writing the review.
- Restaurant Name — Name of the restaurant being reviewed.
- Rating — Numerical rating score given by the reviewer.
- Review Text — Textual content of the review.
- Date of Review — Date when the review was posted.
- Birth Year — Year of birth of the reviewer.
- Marital Status — Marital status of the reviewer (e.g., Single, Married).
- Has Children? — Whether the reviewer has children (Yes/No/Unknown).
- Vegetarian? — Whether the reviewer is a vegetarian (Yes/No/Unknown).
- Weight (lb) — Reported body weight of the reviewer in pounds.
- Height (cm) — Reported height of the reviewer in centimeters.
- Average Amount Spent — Typical spending amount category (e.g., Low, Medium, High).
- Preferred Mode of Transport — Usual transportation method of the reviewer.
- Northwestern Student? — Whether the reviewer is a Northwestern University student.

We initially find that columns *Review Text, Birth Year, Martial Status, Has Children?, Vegetarian? Height (lb), Weight(cm), Average Amount Spent,* and *Preferred Mode of Transport* have missing values. We find that a whopping 1405/1500 of the data points don't specify whether they're vegetarian, so we impute these missing values with "Unknown". We don't do mode imputation, because it would be catastrophic for our recommender system to suggest non-vegetarian restaurants to specifically vegetarian customers. The variable with the second most missing data points is *Review Text*, for which we just imputed the missing values with an empty string, since this variable is crucial for building our recommender system, but we don't want to add any extra words to skew the TF-IDF scores later on for certain words (such as "No"). For the rest of the numerical variables (*Birth Year*, *Weight (lb)*, and *Height(cm)*, we performed median imputation (median instead of mean is safer because of outliers) and created a missing-flag column for each of them, since the missingness of the variables can have some implications for the kinds of people the reviewers are. For the rest of the categorical variables: *Martial Status, Has Children?,* and *Preferred Mode of Transport*, we impute with "Unknown" just as we did for the *Vegetarian?* variable. Lastly, we just did mode imputation for the categorical variable *Average Amount Spent* since only one data point was missing, and added a missing flag variable for this as well.

After all this, we added the new indicator variables:

- Birth Year_missing — Indicator flag for missing birth year data (1 if missing, 0 if present).
- Weight (lb)_missing — Indicator flag for missing weight data (1 if missing, 0 if present).
- Height (cm)_missing — Indicator flag for missing height data (1 if missing, 0 if present).
- Average Amount Spent_missing — Indicator flag for missing spending amount data (1 if missing, 0 if present).

2.1 Univariate Data Analysis

We see that the *Vegetarian?* variable is highly unbalanced with the majority of data points being missing, so we remove this. We can see the distribution of some of the of the categorical variables in 1.

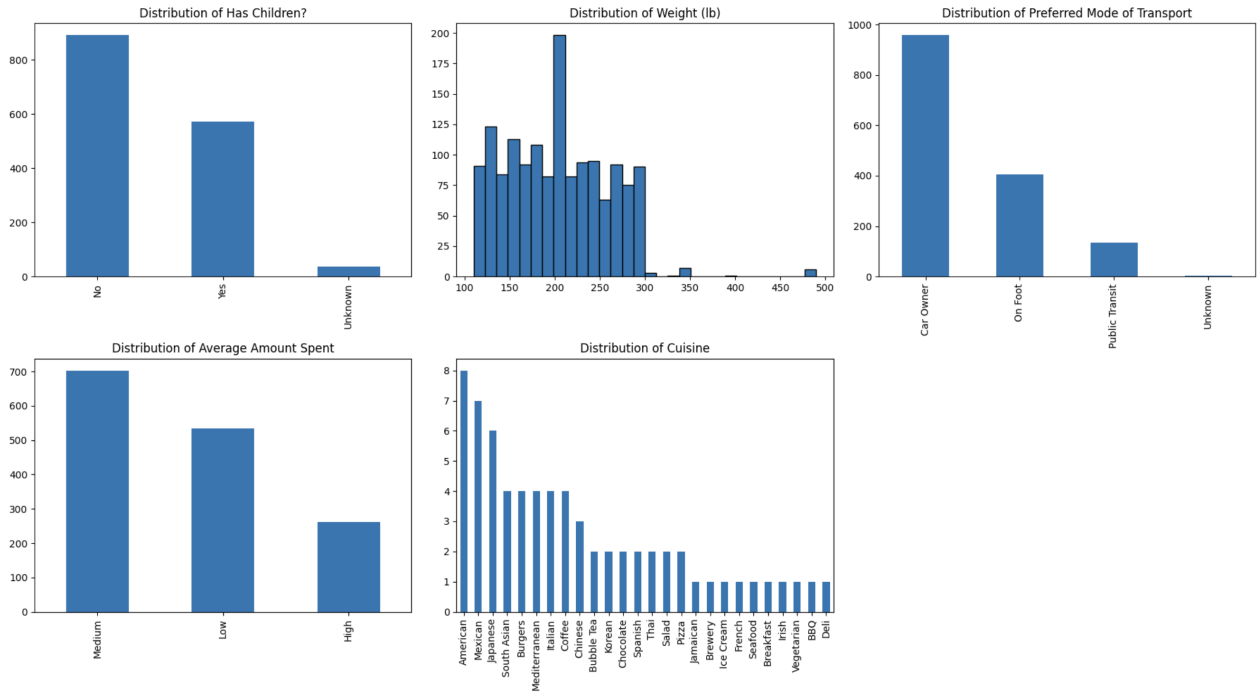


Figure 1: Bar Graphs of Five Categorical Variables

2.2 Ratings by Student Identity (Question 12)

The visualizations in 2 reveal several notable trends in how Northwestern students rate restaurants. In the top left chart, students tend to give slightly higher average ratings than non-students overall, suggesting a marginally more favorable or lenient reviewing behavior. The bottom plot, which breaks ratings down by cuisine, highlights that students consistently rate categories like coffee, chocolate, and pizza more highly than non-students, indicating a possible preference for comfort or casual dining options. Meanwhile, non-students tend to rate cuisines like Mediterranean and salad higher, which could reflect different dietary priorities or age-related preferences. Finally, the top right chart shows restaurant-level differences: Northwestern students give particularly high ratings to places like Union Pizzeria and Tealicious, while non-students rate Kiwins and Kabul House higher. These patterns suggest that Northwestern students gravitate toward accessible, student-friendly venues, while non-students may value different dining experiences, possibly influenced by family-friendliness, portion size, or service expectations.

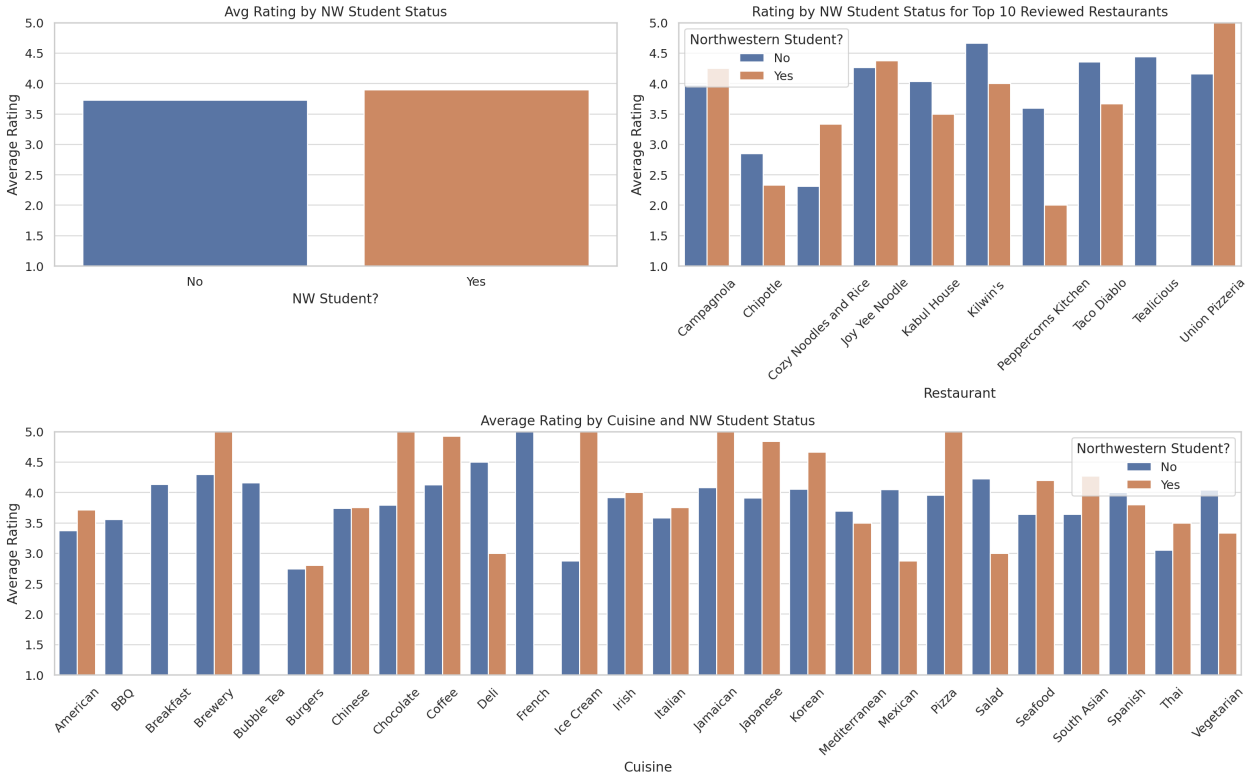


Figure 2: Ratings by Northwestern Student - Yes or No

3 Collaborative Filtering Recommender Systems

We create our first two recommender systems using *user-based collaborative filtering*. We first note that the total number of unique user vectors created equals *1089*. The dimensionality of each vector is *28*.

3.1 Demographic-Based Collaborative Filtering Recommender System

Our first recommender system is a demographic-based collaborative filtering recommender system. To show how it works, we apply it to user Timothy Mace. See 1 for results.

Target User	Most Similar User	Recommended Restaurant(s)	Similar User's Rating
Timothy Mace	Enid Egan	Burger King	5

Table 1: Recommendations for Timothy Mace

To generate a recommendation for Timothy Mace, our system first identifies the user whose feature vector has the smallest Euclidean distance from Mace's — in this case, Enid Egan. It then extracts all restaurants that this most similar user has rated 4 or higher. Since Egan has only rated Burger King at or above this threshold, the system recommends just that one restaurant to Mace.

This outcome highlights a key limitation: the number of recommendations depends on the similar user's review history. To address this, we can revise the system to consider the top k most similar users and/or adjust the minimum rating threshold. These changes would increase the likelihood of generating multiple recommendations for any given user.

3.2 Review-Based Collaborative Filtering Recommender System

Our second user-based recommender system is a review-based collaborative filtering via rating similarity. To show how it works, we apply it on user Sarah Belle. See 2 for results.

Target User	Most Similar User	Recommended Restaurant(s)	Similar User's Rating
Sarah Belle	Steven Johnston	Mumbai Indian Grill	5

Table 2: Recommendations for Sarah Belle

From our example, we can see that this recommender algorithm works by comparing users based on their restaurant review patterns rather than their demographic attributes. Each user is represented as a vector of length equal to the number of restaurants (in this case, 73), where each entry is the user's rating for a restaurant, and missing values are filled with the average rating for that restaurant. Using these vectors, the system computes the Euclidean distance between the target user and all others to identify the most similar user in terms of review behavior. For Sarah Belle, the algorithm found that Steven Johnston had the most similar review pattern. The system then recommended restaurants that Steven rated highly (rating 4), which in this case was only Mumbai Indian Grill, rated 5 stars.

However, this algorithm does not always yield multiple recommendations for every user. If the most similar user has reviewed very few restaurants—or has only rated one or two restaurants with a score of 4 or higher—the system may only return one recommendation or none at all. To address this limitation, the algorithm can be extended in two ways: (1) consider the top k most similar users rather than just one, and aggregate their high-rated restaurants, or (2) lower the rating threshold to include restaurants rated 3 or higher, thereby broadening the pool of potential recommendations. Both strategies would help ensure that more users receive a richer and more diverse set of suggestions.

4 Predicting Restaurant Ratings

4.1 Predictors: Cuisine Type and User Demographics

4.1.1 Using Linear Regression

We then sought to compare the performance of linear and logistic regression models in predicting restaurant ratings, with demographic data and cuisine type as the predictors. To do this, we first split the dataset into a training and a testing dataset and fit a linear and logistic regression separately on the training dataset and evaluated the two models using the test dataset. We see that the error for the linear regression model was large, with an **MSE of 1.95**, which means that the predicted ratings are off about 1.4 stars. This is a fairly large number given that we are rating restaurants on a scale from 1 to 5. This warrants further analysis: maybe there are several huge outliers, which increases MSE. We visualize the results in 3, which plots predicted ratings against actual ratings. The dotted red line shows where the plots should ideally land. It is clear that our linear regression model has very poor predictive power for low actual ratings.

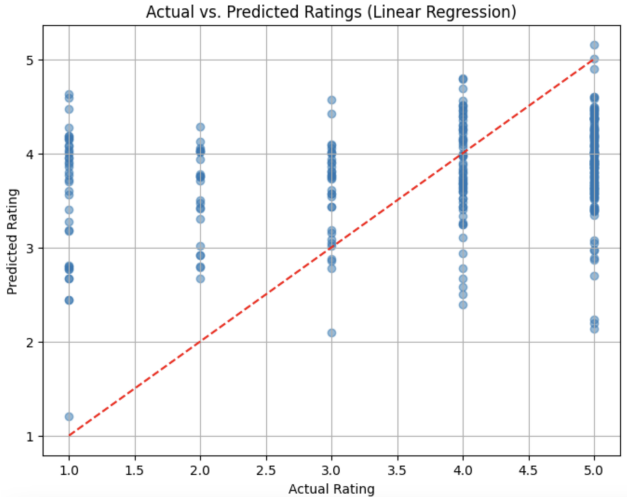


Figure 3: Actual vs Predicted (using Linear Regression) Ratings of all Restaurants in Evanston

4.1.2 Using Logistic Regression

On the other hand, we see that the logistic regression had an overall **71 percent accuracy rate** for predicting the category to which a review belonged to (good or bad). We get that the "good" class has a recall rate (defined as how accurately we predict good reviews) of 91%, which is very high. However, the "bad" class has a recall rate of only 30%. This implies that our model has a huge bias in favor of positive reviews, which is worrisome in its ability to extrapolate. Furthermore, we find that the good class has a precision rate (defined as the precentage of predicted "good" reviews that are actually in the good category) of 73%. The bad class has a precision rate of 60%. The precision rates for both classes is decently high. Overall, we get that the logistic regression is decent in prediction, and definitely better than the linear regression model.

4.1.3 Testing Linear and Logistic Regression on Sample Review

For our sampled review, we find that the true rating is 5 (in the good rating category), but the linear regression model predicts 2.87 and the logistic regression predicts that this review is in the bad category. In this example, we find that both models did very poorly in predicting the accuracy of the rating.

4.1.4 Lasso Regularization on Linear Regression Model

Adding an L1 penalty to the linear regression model significantly simplified the model by reducing the number of selected features from many to just one: Birth Year. While the standard linear regression model achieved a mean

squared error (MSE) of 1.95, the Lasso model slightly underperformed with an **MSE of 2.03**, indicating a small trade-off in predictive accuracy for improved interpretability. The only feature with a non-zero weight was Birth Year, with a small positive coefficient, suggesting that younger or older users may systematically give higher or lower ratings. All other features, including cuisine type and other demographic attributes, were shrunk to zero, meaning they had negligible or redundant predictive value under regularization. This highlights how L1 regularization can expose the most influential variables and eliminate noise, albeit sometimes at the cost of slightly reduced performance.

4.2 Predictor: Embedded Reviews (Question 11)

We also wanted to build a model to predict ratings of Evanston restaurants using the text reviews. We accomplished this by embedding the reviews using Sentence Transformers and using these embeddings to train a logistic regression model to predict the review sentiment as positive or negative. In training, we let reviews with ratings four or better to have positive sentiment, and all else to have negative sentiment.

We find that this model has an **MSE of 0.14** and an **overall accuracy of 0.80**. This MSE is significantly lower than that of the linear regression model that used cuisine type and demographic data as predictors, which was 1.95, and that of the lasso regression model (with same predictors as those of linear regression), which was 2.03. This reveals that learning the reviews is a more accurate way of predicting whether the rating will be good or bad than using demographic data and cuisine type.

5 Predicting Coffee Shop Ratings

5.1 Predictor: User Demographics

We then want to observe which demographic features are the best and worst predictors of ratings, *specifically* for coffee shops in Evanston. We note that there are only four coffee shops in our dataset: "Peet's Coffee," "Pâtisserie Coralie," "Brothers K Coffee," and "Philz Coffee." We find that fitting a linear regression model, with the ratings of these three coffee shops as the response and demographic features as predictors, yields an **MSE of 2.77**. This means that my ratings are off by about $\sqrt{2.77} \approx 1.66$ rating points, which is a significant error margin given we are rating from a scale of 1-5. We can visualize this in 4.

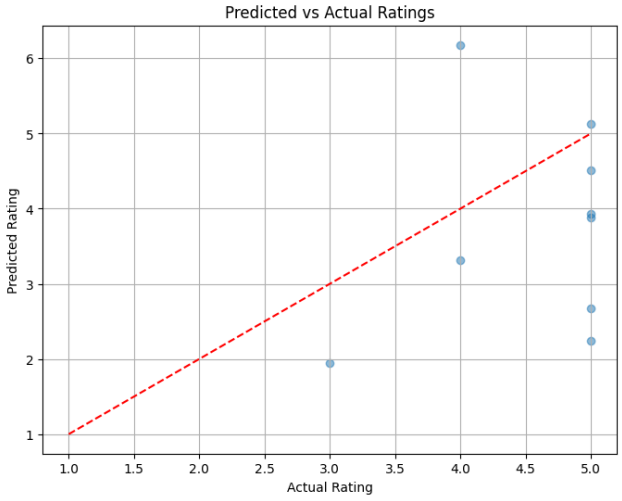


Figure 4: Predicted vs Actual Ratings of Coffee Shops in Evanston

Hence, we can see that the linear model poorly predicted the ratings of coffee shops. To explore potential causes of the poor predictive power, we began by observing whether feature selection was the issue. We found that the demographic features that had the highest positive coefficients were

1. *Vegetarian? = Yes* (Coeff = 1.93)

2. *Preferred Mode of Transport=Public Transport or Unknown* (Coeff = 1.40)

3. *Northwestern Student? = Yes* (Coeff = 1.23)

and those with the most negative coefficients were

1. *Average Amount Spent = Low* (Coeff = -1.34)

2. *Average Amount Spent?=Medium* (Coeff = -0.79)

3. *Martial Status = Single* (Coeff = -0.59)

From these weights, we can infer that Northwestern students are very likely to visit coffee shops. This makes sense because students may prefer to study in coffee shops and/or caffeinate themselves throughout the day to stay productive. This may also be because the majority of the population surrounding downtown Evanston consists of Northwestern students. Further, we can infer from the model that vegetarians are very likely to highly rate coffee shops. This may be because vegetarians are more likely to follow health-conscious, eco-conscious, or alternative lifestyles. These lifestyles often align with the culture of third-wave coffee shops, which emphasize: organic or plant-based milk options, sustainable practices, and/or community-oriented spaces. Also, it may be the case that vegetarians are often younger and more urban — demographics that also frequent and rate coffee shops highly.

On the other hand, we see that those who spend a low or medium amount are the least likely to give high ratings to coffee shops. This may possibly be because the more expensive on the menus taste better, or those who complement their drink with food enjoy a more pleasant and complete culinary experience.

We are concerned that *Preferred Mode of Transport=Public Transport or Unknown* is one of the best predictors of coffee ratings, despite there only three data points falling under this category. This might mean the model may be learning from noise rather than a meaningful pattern. We want to warn anyone who wants to extrapolate our model to other coffee shops to proceed with extreme caution and consider removing these three data points before extrapolating. However, since we do not have many data points to begin with, for the purposes of this paper, we will not be removing these data points in our analysis.

We are also concerned that demographics are not the best predictors of how highly customers rank coffee shops. If this were true, then that would explain the high MSE value we see from our model. We believe that there may be more important predictors of how people rate coffee shops, such as the location of the store, the ambience, the diversity of the menu, etc. In other words, the traits of the stores themselves are probably better indicators of how highly customers will rank the coffee stores. This is, however, beyond the scope of our analysis - we invite those interested in answering this question to further explore it.

6 Conclusion

In this project, we developed and evaluated multiple recommender systems and predictive models using a rich dataset of restaurant reviews and demographic information from Evanston, IL. Our collaborative filtering systems explored two approaches: one based on demographic similarity and another based on review similarity. While both systems could make user-personalized recommendations, they were often limited by the sparsity of high-rated reviews from similar users. We also built predictive models using linear and logistic regression with user demographics and cuisine types as features. These models showed limited predictive power, particularly in predicting low ratings, with the linear regression model producing a relatively high mean squared error (MSE) of 1.95. In contrast, a model trained on review text embeddings significantly outperformed these approaches, achieving an MSE of just 0.14 and an accuracy of 80%, suggesting that textual content holds richer information than demographic or categorical inputs.

We also examined a narrower use case—predicting coffee shop ratings using demographic features—and found similarly weak performance, with a high MSE of 2.77. Although some features, such as being a Northwestern student or identifying as vegetarian, were associated with higher coffee shop ratings, we cautioned against over-interpreting these findings due to the small sample sizes and potential confounding variables. For example, one of the top predictors—"Preferred Mode of Transport = Public Transit or Unknown"—had only three supporting data points, suggesting a risk of the model capturing noise rather than signal. Overall, while our models uncovered interesting trends (e.g., student preferences for comfort-focused establishments), they also highlighted the limits of shallow demographic features in modeling human taste, emphasizing the importance of richer input data, like text or contextual information, in building robust recommendation systems.